Zihan Zhou

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EDUCATION BACKGROUND

The Chinese University of Hong Kong, Shenzhen (CUHK-SZ), China 09/2019 – 06/2023

B.S. in Mathematics and Applied Mathematics | Cum GPA: 3.551/4.0 | 2019-20 Dean's List and 2021-22 Dean's List

Core Courses: Multivariable Calculus | Advanced Linear Algebra | Real Analysis | Probability Theory | Mathematical Modeling | Optimization | Partial Differential Equations | Discrete Mathematics | Numerical Analysis | Stochastic Processes | Reinforcement Learning | AI in Medical Imaging | Programming Paradigm | Data Structures

SKILLS

Programming Languages: C/C++ | Java | Python

Libraries: TensorFlow | NumPy | fastai | SciPy | PyTorch | XGBoost | Pandas | Seaborn | Scikit-Learn | Matplotlib

EXTRACURRICULAR ACTIVITIES

***Team Leader* | Mathematical Contest in Modeling (MCM)** 02/2022

- Built, trained, and validated an LSTM model to predict the price trend of U.S. dollars, gold, and bitcoins, garnering the Meritorious Award (Top 6%):
  - Established a prediction model based on 1) signals from the Moving Average Convergence Divergence (MACD) strategy, 2) quantification of market tendency and volatility, 3) training and validation of an LSTM model with hyperparameter tuning via cross-validation and grid search, and 4) an ARIMA model for online handling of time series data.
  - Proved through stochastic perturbation and empirical simulation the optimality of the developed strategy.
  - Explored the correlation between transaction cost and frequency; boosted the model robustness with a new voting system.

***Team Leader* | The "Mathor Cup" China Collegiate Mathematical Modeling Challenge** 12/2021 – 01/2022

- Built and validated a data-driven workflow for predicting used car prices and identifying key factors driving the sales volume:
  - Collected real-world data, followed by data cleansing and feature engineering (one-hot encoding, feature hashing) to ensure validity, accuracy, completeness, consistency, and uniformity of the dataset; performed data visualization and exploratory data analysis to identify anomalies, test hypotheses, and extract actionable patterns.
  - Collectively trained and validated random forest, SVM, neural network, XGBoost, and LightGBM models to drive the price prediction; enabled model stacking via kernel ridge regression to boost predictive accuracy.

Kaggle Challenge: Pawpularity Contest 09/2021 – 12/2021

- Employed EfficientNetB0, Swin-Transformer, and ConvNet models in fastai library to analyze pet photos, with a focus on determining a photo's appeal and suggesting an improvement to enhance the chance of rescue animal adoption:
  - Performed training set cleansing and removed anomalies; visualized image feature distribution; final score ranked top 7% globally (249/3538)

***Team Leader* | China Undergraduate Mathematical Contest in Modeling (CUMCM)** 09/2021

- Built a data-driven model in Python to characterize and optimize a process for producing C4 alkene from ethanol:
  - Modeled chemical reaction dynamics with a set of ordinary differential equations, then solved the equations using the 4th-order Runge-Kutta method while attaining a balance between numerical stability and computational efficiency.
  - Calibrated the model against experimental data via the least squares method, maximum likelihood estimation, and multiple-iterative optimization, followed by optimizing the reaction condition (catalyst selection, temperature).

***Vice President* | CUHK-SZ Badminton Club** 10/2019 – 10/2020

- Planned and coordinated multiple on-campus events to promote badminton activities in the student community, such as intramurals, tournaments, and contests; managed new member recruitment and various logistic items; organized weekly training and a series of teambuilding events; significantly bolstered leadership, interpersonal, and multitasking skills.

RESEARCH EXPERIENCES

Research Assistant | Combinatorial Optimization Group, CUHK-SZ 10/2021 – Present

- Propose a sequential learning paradigm by decoupling a complex system for handling irregularly sampled and clustered sequential observations:
  - Establish a deeply coupled differential equation set based on a decoupling mechanism that raises not only each latent sub-system but also another system capturing the evolving interactions between the sub-systems.
  - Model the system of interactions by a projected differential equation, followed by a generalization within the context of Bregman Divergence.
  - Experimentally validate the new approach using synthetic and real-world datasets, demonstrating its advantage in handling complex and clustered sequential data compared to the state-of-the-art.
  - This work has been summarized and submitted as a first-authored regular research paper to ICLR2023.
- Apply the sequential learning model to video decoupling and compression in collaboration with China's Huawei.

Infant Cry Recognition Based on Deep Learning
The 2nd "Shenchuang Cup" International Collegiate Innovation and Entrepreneurship Competition 2022

- Employed CNN and LSTM models to enable automated information extraction from infant cry, covering a wide range of needs such as food, water, diaper change, pampering, and medical treatment.
  - Applied continuous wavelet transform (CWT) and short-time Fourier transform (SIFT) for feature extraction to prepare input to a convolutional neural network for classification.
  - Leveraged Mel Frequency Cepstral Coefficient to represent the power spectrum of the cry sound.
  - Experimentally demonstrated the system's effectiveness, flexibility, and robustness in handling various conditions; completed a systematic comparison to identify the optimal methods under various conditions.
  - Delivered a detailed plan for commercializing the product; performed profit model design, investment and financing planning, market analysis, SWOT analysis, marketing model analysis, financial analysis, and risk estimation.

Literature Review on Meta-Reinforcement Learning Towards Efficient Sampling 03/2022 – 05/2022

- Completed an extensive literature review on a set of meta-reinforcement learning (RL) algorithms:
  - Focused on analyzing and comparing the pros and cons of each algorithm in terms of adapting efficiently to different tasks under a common distribution to eliminate the need for extensive sampling and trials.
  - Delivered a comprehensive review manuscript to document the findings and insights.

Crop Health Monitoring via Computer Vision and Pattern Recognition
***Research Assistant* | CHUK-SZ AIoT Precision Agriculture Program** 05/2021 – 10/2021

- Designed an urban farming system, which, based on computer vision and artificial intelligence, can automatically monitor crop health and prevent pest infestations; key components include:
  - Collected 1K+ images of the infested crop from local farmland; performed format reconciliation, data augmentation (shuffle, rotation, scaling), and identified regions of interest (ROIs).
  - Built, trained, and validated VGG-19, ResNet-34, ResNet-50, ResNet-101, and GoogLeNet models to enable real-time pest detection, attaining an overall accuracy of 89.8%; demonstrated the algorithms' superiority to the conventional CNN models in terms of speed and accuracy.
  - Enabled iterative update of network weights with the Adam optimizer; controlled the learning rate using the cosine annealing algorithm.
  - Optimized the developed models, covering multi-leaf detection within a frame, improvement of mean average precision, and the expansion of crop/pest categories.

Online Summer Research Programme, University of Cambridge 07/2021 – 08/2021

- Designed, implemented, and evaluated a set of computational algorithms for computing p->q matrix norm, which generalizes the spectral norm of a matrix and the Grothendieck problem with wide real-world applications:
  - Performed an extensive literature review on the theory and applications of matrix norm in quantum mechanics and machine learning; systematically implemented and tested n-sphere, critical point, gradient descent, and multiple gradient descent algorithms
  - Employed probability theory and matrix algebra to analyze the time complexity, convergence rate, and error bound for each algorithm; theoretically evaluated the upper and lower bounds of the norm, followed by evaluating the global optimality of the algorithms; enabled visualization of matrix norm distribution to facilitate the study.

WORK EXPERIENCES

***Algorithm Engineer Intern* | Beijing ByteDance Technology Co., Ltd., China** 06/2022 – 07/2022

- Implemented, tested, and deployed a set of natural language processing algorithms and deep learning models to gauge the popularity of bloggers via online data mining:
  - Constructed a transformer architecture to enable text classification, focusing on differentiating positive and negative reviews based on input sentences, attaining 92% overall accuracy.
- Analyzed and ranked academic publications within a specified domain based on abstracts and hot words:
  - Conducted keywords extraction with TF-IDF statistics and latent semantic analysis; clustered semantic vectors with cosine similarity for summarizing semantic themes

COURSE PROJECTS

***AI in Medical Imaging* Course Project, CUHK-SZ** 04/2022 – 05/2022

- **Neural Controlled Differential Equations for Action Localization and Keyframes Extraction**
  - Led a team of three to implement a framework for using the neural controlled differential equations model to enable action localization and keyframe extraction:
    - Clustered the video frames of upper endoscopy based on hidden state feature vectors for video segmentation, followed by identifying keyframes best-representing segments to inform the diagnosis.
- **Lung Disease Diagnosis Using X-Ray Imaging and Machine Learning** 03/2022 – 04/2022
  - Trained an XGBoost model to drive lung disease diagnosis based on X-ray images, gauged the model performance using AUC, and demonstrated its superiority to deep learning classifiers (ResNet18) in overfitting issues.
  - Used GradCAM++ methods to visualize and identify the key input image segments dictating the model performance.

PUBLICATION

Learning to decouple complex system, Zihan Zhou, Tianshu Yu, accepted by ICML2023